BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 091 · Lesson 091 · Fetch–Decode–Execute: A Register Trace

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which register holds the current instruction?

Q2. What is the role of MAR?

Q3. Where is the next instruction address kept?

Q4. A full adder has A=1,B=0 and carry-in=1. State Sum and carry-out.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

fetch — Retrieve next instruction.

decode — CU interprets instruction.

execute — Carry out operation.

CIR — Current Instruction Register.

ACC — Accumulator.

New learning

Observe — Fetch copies the address from PC to MAR; memory read transfers the instruction through MDR into CIR. In a sequential fetch, PC advances.

Explain — The Control Unit decodes the current instruction held in CIR and produces the necessary control signals.

Evaluate — Execution of LDA 80 reads the data held at memory address 80 into ACC; the next instruction is not executed until its own fetch.

Worked example

PC=30, memory[30]=ADD 70, memory[70]=5, ACC=4. Trace after fetch and execute.

1. Explain the mechanism — Fetch: MAR30, CIR ADD70, PC31. Execution reads memory[70]=5 and ALU adds to ACC4, giving ACC9. — The PC increments in fetch while the ALU operates on ACC and fetched operand.

We Do · Practise with guidance

PC=42, memory[42]=SUB 90, memory[90]=3, ACC=11. Trace registers and result.

Explain the mechanism

After fetch PC43,CIR SUB90. Operand read from90 into MDR=3; ALU computes11−3, ACC8.

The addressed operand must not be confused with instruction address42.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace the fetch of instruction at PC20 into MAR,MDR,CIR; give PC after sequential increment. [6]

Question 2

Explain difference between decode and execute for LDA 80 when memory[80]=13. [6]

Question 3

An answer states 'MDR stores address 80 and MAR stores the fetched 13'. Correct the error with a precise memory-read explanation. [6]

Exit Ticket · Answer before checking

During a sequential fetch, MDR stores the instruction transferred from memory rather than its address.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. CIR.

Q2. Stores address being accessed.

Q3. PC.

Q4. $1+0+1=10_2$, so Sum0, carry-out1.

You Do pages · Independent questions

Q1. MAR20 [2], memory[20] via MDR [2], CIR gets instruction [1], PC21 [1].

Q2. CU interprets opcode/operand [2], memory operand address80 [1], data13 via MDR [1], ACC13 [2].

Q3. MAR80 carries target address [2], memory sends value13 on data path [2], MDR receives13 [2].

Exit Ticket · Model answer

True. MAR holds the address; MDR the returned data/instruction.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: